

5 C++ implementation

5.1 Sequential baseline

The deterministic model above is also the reference problem for the performance analysis of this chapter. Timings are taken in a separate release build with visualisation and diagnostic output disabled, thread counts pinned to one, and reported as the median of five runs after one warm-up.

Elements per connection	Unknowns	Matrix nonzeros	100 steps [s]	Per step [s]
1	83	2343	0.0832	0.00083
2	1213	5733	1.8694	0.01869
4	3473	12513	1.5486	0.01549
8	7993	26073	2.2343	0.02234

Table 6: Sequential baseline for the deterministic model. Assembly accounts for less than one percent of the total in every refined case.

Two facts emerge from Table 6 and they set the agenda for what follows. First, in every refined case more than 99% of the time is spent in the time loop (95.9% at one element per connection) and almost none in assembly or input, so the target for optimisation is the repeated sparse factorisation forced by the semi-implicit scheme (Table 5), not the finite element assembly. Second, the cost per step is not a function of the number of unknowns alone: the two-element mesh, with 1213 unknowns, costs more than the four-element mesh with 3473. The two-element mesh inherits a hub-dominated sparsity pattern from the connectome, and sparse factorisation depends on the elimination ordering and the resulting fill-in at least as much as on the matrix dimension.

For scale, [9] report that their nodal model runs its 100 implicit steps in 0.55 s without output on a standard laptop; our metric-graph formulation at the same 83 unknowns takes 0.083 s for its 100 semi-implicit steps. The machines and the schemes differ, so this is an order-of-magnitude check rather than a comparison, but it confirms that the P1 metric-graph formulation carries no cost penalty over the nodal network model at the coarsest mesh, while allowing arbitrary refinement inside the connections.

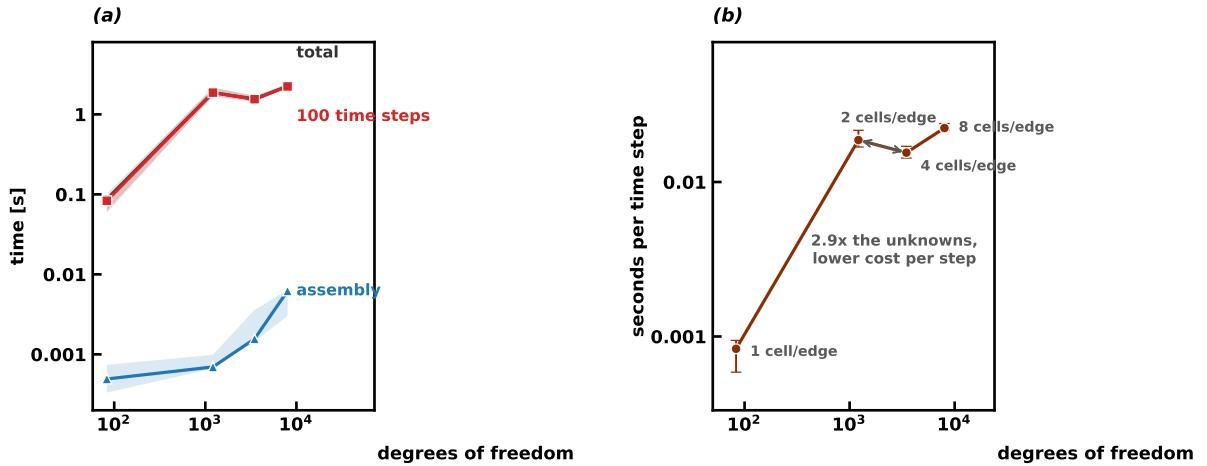


Figure 28: Sequential baseline. (a) assembly, time loop and total against the number of unknowns, with the band spanning the minimum and maximum of five runs. (b) cost per time step, showing that it is not a function of the unknown count alone: the two-element mesh costs more per step than the four-element one.

This is precisely the observation that motivates the reordering study of the next section, where bandwidth, profile, factor fill and factorisation time are measured for different orderings before any parallel scaling result is interpreted.

5.2 Matrix ordering and the sequential solver

With the semi-implicit scheme of Table 5 every time step advances the concentration by solving one linear system. Writing M for the mass matrix, H for the discrete Hamiltonian of Section 2.2 and $W(v)$ for the reaction matrix assembled with the weight $\alpha(1 - v)$, the Crank–Nicolson update with the second-order extrapolation $\hat{c}^{n+1} = \frac{3}{2}c^n - \frac{1}{2}c^{n-1}$ reads

$$K^n c^{n+1} = b^n, \quad K^n = M + \frac{\Delta t}{2}H - \frac{\Delta t}{2}W(\hat{c}^{n+1}), \quad (13)$$

with $b^n = (M - \frac{\Delta t}{2}H + \frac{\Delta t}{2}W(\hat{c}^{n+1}))c^n$. The sparsity pattern of K^n is that of $M + H$ and never changes; its values change at every step through the extrapolated reaction weight, so the solver must refactorise at every step. All three matrices are symmetric, hence so is K^n . The solver used in the preceding chapters treats the system with the sparse LU factorisation of Eigen and its default fill-reducing column ordering, COLAMD, recomputing the factorisation from scratch at each step. This section asks whether that factorisation, which Table 6 identifies as the dominant cost, is performed in the right ordering.

To compare orderings under the exact conditions of the production run we replay the same time loop, with the same matrices and right-hand sides, and hand the matrix of every step to one of six variants: the production LU with COLAMD, LU on the natural ordering, LU with the approximate minimum degree ordering AMD, LU on a reverse Cuthill–McKee (RCM) permutation of the matrix and, exploiting symmetry, the LDL^T factorisation on the AMD and on the natural ordering. Over one hundred steps the trajectories of all variants agree with the production one to within 4×10^{-14} in the maximum norm, so the comparison is between numerically equivalent solvers. For each variant we record the bandwidth and profile of the matrix handed to the solver, the number of nonzeros of its factors and the per-step cost split into the rebuild of the matrix, the factorisation and the triangular solves.

The natural ordering is not an arbitrary baseline, because the assembly numbers the unknowns in a specific way: the interior unknowns of each connection are numbered consecutively, connection after connection, and the 83 vertex unknowns come last. In this numbering K^n is an arrow matrix, a block-diagonal part with one tridiagonal block per connection bordered by the rows and columns of the vertices. Gaussian elimination in natural order eliminates the interiors of each connection inside their own tridiagonal block and produces new nonzeros only in the border, so the fill-in is essentially confined to an 83×83 block regardless of refinement.

Ordering	2 elements, $n = 1213$		8 elements, $n = 7993$		64 elements, $n = 71273$	
	factor	nonzeros	step [ms]	factor	nonzeros	step [ms]
LU, COLAMD	182208	26.84	104934	21.16	484614	140.31
LU, natural	10522	2.35	51202	10.43	430882	81.38
LU, RCM	80570	8.13	1103850	162.93	11594218	2013.1
LDLT, natural	5261	0.61	25601	1.93	215441	21.04

Table 7: Ordering study on the connectome meshes with 2, 8 and 64 elements per connection: nonzeros of the computed factors and median per-step cost (matrix rebuild, factorisation and triangular solves) over 100 steps of the semi-implicit scheme. RCM is applied as an explicit symmetric permutation of the matrix; the other variants reorder internally.

Table 7 and Figure 29 collect the results. Three observations follow. First, the production ordering is responsible for the anomaly of Table 6: on the two-element mesh COLAMD produces 182208 factor nonzeros, more than the 77828 it produces on the four-element mesh, so the smaller problem really is more expensive to factorise. The anomaly is a property of the ordering heuristic on this hub-dominated pattern, not of the problem. Second, minimising bandwidth is the wrong objective for this structure. RCM reduces the bandwidth from 7989 to 937 on the eight-element mesh, but it destroys the arrow structure: at 64 elements per connection its profile is eleven times that of the natural ordering and its factors carry 27 times the nonzeros, for a step of about two seconds. Third, the natural ordering is already close to fill-optimal, below COLAMD

on every refined mesh, and the symmetric LDL^T factorisation on the natural ordering halves the remaining work: at 64 elements per connection it stores 215441 factor nonzeros against the 484614 of the production configuration and completes a step in 21.0 ms against 140.3 ms, a factor of 6.7. AMD paired with LDL^T finds essentially the same fill, 214939 nonzeros, but repeats its ordering analysis at every rebuild and ends up nearly twice as slow, so the natural ordering obtains the same quality at no cost.

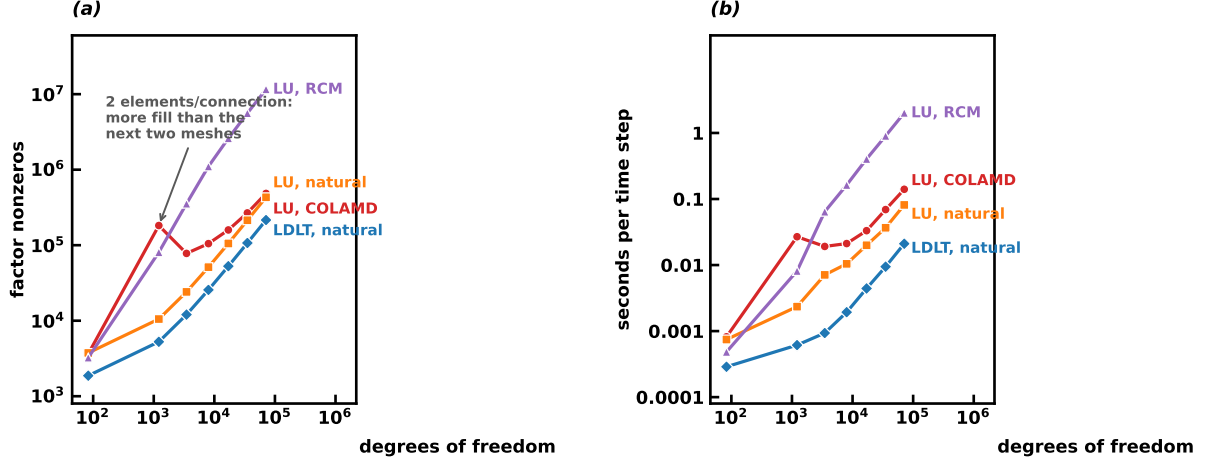


Figure 29: The ordering study. (a) factor nonzeros against the number of unknowns for the four orderings of Table 7; the marked point is the two-element mesh, on which the production ordering fills more than on the next two meshes. (b) median cost per time step. The LDL^T factorisation on the natural ordering is the sequential reference adopted for the rest of the chapter.

We therefore adopt the LDL^T factorisation on the natural ordering as the sequential reference. This is an algorithmic improvement, obtained with the same discretisation and the same arithmetic. It also moves the bottleneck: at 64 elements per connection the factorisation now costs 6.1 ms and the triangular solves 0.9 ms, while rebuilding the sparse matrix costs 14.0 ms per step. Once the ordering is right, most of the step is spent assembling a global matrix whose structure repeats, connection by connection, a pattern that the factorisation then eliminates. The next section removes the global matrix altogether.

5.3 Static condensation

The arrow structure can be exploited before any global factorisation is performed. Let E collect the interior unknowns of the connections and V the 83 vertex unknowns; splitting the system of Eq. (13) accordingly gives

$$\begin{pmatrix} K_{EE} & K_{EV} \\ K_{VE} & K_{VV} \end{pmatrix} \begin{pmatrix} c_E \\ c_V \end{pmatrix} = \begin{pmatrix} b_E \\ b_V \end{pmatrix}, \quad K_{EV} = K_{VE}^\top, \quad (14)$$

where K_{EE} is block diagonal with one tridiagonal block per connection, because interior nodes couple only to their neighbours on the same connection. Eliminating c_E exactly gives the Schur complement system on the vertices,

$$S c_V = g, \quad S = K_{VV} - K_{VE} K_{EE}^{-1} K_{EV}, \quad g = b_V - K_{VE} K_{EE}^{-1} b_E, \quad (15)$$

followed by the back substitution $c_E = K_{EE}^{-1}(b_E - K_{EV} c_V)$. No approximation is involved: the pair (c_E, c_V) solves Eq. (13) exactly, and the reduction is Gaussian elimination in the natural ordering, written so that the elimination of each connection becomes an independent task.

The per-connection work is small and structured. A connection with m elements contributes a tridiagonal $(m-1) \times (m-1)$ block of K_{EE} , two coupling columns towards its endpoints and

the interior part of the right-hand side; the entries are the same element integrals as in the production assembly, evaluated with the extrapolated reaction weight of the current step. The Thomas algorithm factorises the tridiagonal block in $O(m)$ operations, and sweeps against the right-hand side and the two coupling columns produce the 2×2 contribution of the connection to S , its two entries of g and the vectors reused by the back substitution. The interface matrix S is 83×83 with the sparsity of the connectome plus the diagonal; we store it as a dense matrix and factorise it by a dense LDL^T at every step, a constant cost of about 0.1 ms independent of refinement. The global sparse matrix of Eq. (13) is never assembled.

For the timing study the connectome mesh is refined up to 128 elements per connection, that is 143593 unknowns. The topology is kept fixed and only the finite-element resolution along each connection is increased, providing a controlled increase in the number of unknowns without modifying the underlying connectome.

Elements per connection	Unknowns	Full system [ms]	Condensed [ms]	Ratio
8	7993	3.61	1.03	3.5
16	17033	8.34	1.99	4.2
32	35113	13.52	4.11	3.3
64	71273	35.02	6.73	5.2
128	143593	64.45	10.99	5.9

Table 8: Static condensation against the full-system LDL^T factorisation on the natural ordering: median per-step cost over 100 steps. The two solvers advance the same trajectory back to back in the same process, so each ratio compares two measurements of the same run.

Table 8 compares the condensed step with the full-system LDL^T reference of Section 5.2; the two solvers run back to back in the same process, so each ratio compares two measurements of the same run, and the absolute times differ from those of Table 7 by the run-to-run variability of the machine. The condensed step is between 3.5 and 5.9 times faster than the full-system one, with the largest gain on the finest mesh, and its cost grows roughly linearly with the number of unknowns. Figure 30(b) shows where the time goes: at 128 elements per connection the per-connection elimination takes 10.56 ms, the back substitution 0.34 ms and the interface solve 0.09 ms, less than one percent of the step. Almost the entire step is independent work on the connections, which is the property the parallel versions of the next section exploit.

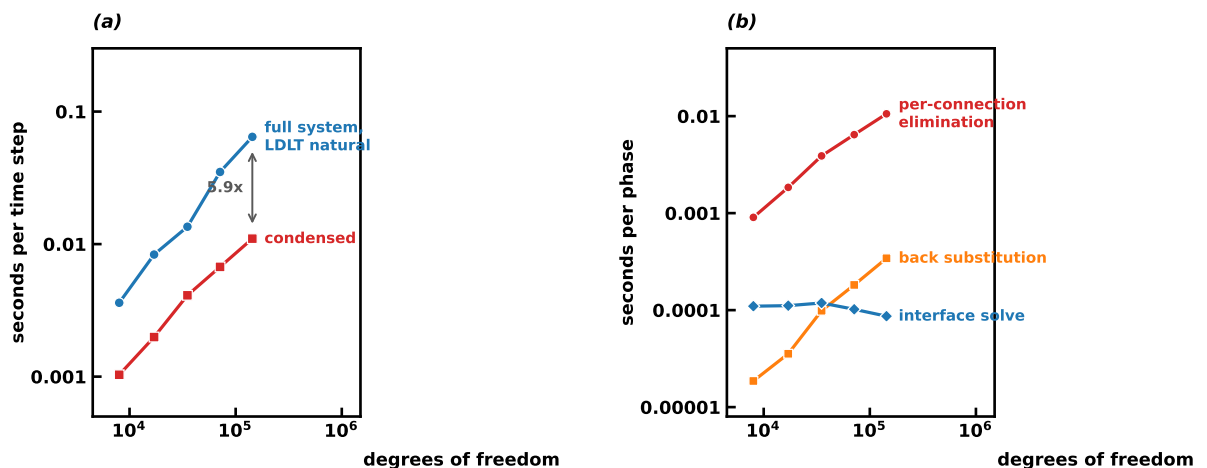


Figure 30: Static condensation. (a) median per-step cost of the condensed solver against the full-system reference, with the ratio at the finest mesh. (b) phase split of the condensed step: the interface solve is constant across refinement and the step is dominated by the independent per-connection eliminations.

The condensed trajectory was compared step by step with the production solver over one hundred steps on the meshes up to eight elements per connection, with maximum differences

of about 10^{-14} , and against the full-system LDL^T solver on the finer meshes, with differences below 3×10^{-14} : static condensation is the same computation reorganised. Taking the two experiments together, the per-step cost at 64 elements per connection falls from the 140.3 ms of the production configuration to 6.73 ms, a factor of about twenty from ordering, symmetry and condensation combined. This is an algorithmic gain, measured with one thread; we keep it separate from the parallel speedups of Section 5.5, which are measured against the condensed solver itself.

5.4 Parallel implementation

Both parallel versions parallelise the condensed step. The natural unit of work is the connection: its elimination and its back substitution touch only its own interior storage and contribute to the interface system through a 2×2 block and two right-hand-side entries. The interface solve stays serial in the OpenMP version and replicated in the MPI version, a choice justified by its constant cost of about 0.1 ms.

5.4.1 OpenMP

The elimination loop and the back-substitution loop over the connections are shared among the threads. Each connection writes to its own slice of the preallocated work arrays, so the loops need no synchronisation; the contributions to S and g are accumulated in thread-private storage and merged once per thread at the end of the elimination loop. A single thread then factorises the interface system and the back substitution runs again in parallel. The arithmetic is identical to the sequential condensed step except for the order in which the per-thread contributions are summed into the interface system, so the two versions agree to round-off: advancing both engines in lockstep and comparing every step gives maximum differences below 3×10^{-14} on the eight-element mesh with two, four and eight threads and on the 128-element mesh with eight threads.

5.4.2 MPI

The connections are assigned to the ranks in contiguous blocks, sized so that every rank owns nearly the same number of interior unknowns. Each rank stores the replicated vertex states and the interiors of its own connections only, eliminates its connections and accumulates its local contribution to the interface system. One reduction phase per step, two `MPI_Allreduce` calls that sum the 83×83 interface matrix and its right-hand side across the ranks, carries a fixed payload of about 55 kB independent of refinement; every rank then factorises the interface system redundantly, which at this size is cheaper than distributing the factorisation, and back-substitutes its own connections without further communication. Per-step communication is therefore constant while the distributed work grows with the mesh. The same lockstep comparison against the sequential condensed solver, gathering the distributed state at every step, gives maximum differences below 10^{-14} on the eight-element mesh with two, four and eight ranks and on the 128-element mesh with eight ranks.

A hybrid variant is obtained by compiling the MPI version with OpenMP enabled, so that each rank runs its elimination and back-substitution loops with threads and ranks and threads can be traded at a fixed number of workers. Its lockstep differences are at the same level.

5.5 Parallel performance

All scaling experiments run on the mesh with 128 elements per connection and 143593 unknowns, chosen because the coarser meshes complete a condensed step in a few milliseconds at most, leaving little work to distribute. The machine is the same laptop as before, with four physical cores and two hardware threads per core (simultaneous multithreading, SMT); up to four workers

each worker is bound to its own physical core and eight workers use the hardware threads. Every configuration advances 200 steps and every reported time is the median over the steps and over three repeated runs. Each experiment measures its own sequential reference in its own session; the references are 9.58, 8.07 and 8.40 ms per step for the OpenMP, MPI and hybrid experiments, a spread that reflects the run-to-run variability of the machine, and every speedup is computed against the reference of its own experiment.

Threads	Per step [ms]	Speedup	Efficiency
sequential	9.58	–	–
2	5.65	1.69	0.85
4	3.37	2.84	0.71
8	2.06	4.64	0.58

Table 9: OpenMP strong scaling of the condensed step on the mesh with 128 elements per connection: medians over 200 steps and three runs, speedups against the sequential condensed reference of the same experiment.

Table 9 reports the OpenMP scaling. Running the OpenMP build on a single thread costs 10.70 ms against the 9.58 ms reference, an overhead of about twelve percent for the threaded code path. From there the scaling is regular: 1.69 at two threads, 2.84 at four and 4.64 at eight. The four-thread configuration is the clean measurement of the physical machine, one thread per core, with a parallel efficiency of 0.71; at eight threads the step still shortens, from 3.37 to 2.06 ms, but each core now runs two hardware threads. The lower efficiency at eight threads is consistent with the use of SMT on a four-core processor and with the increasing relative cost of serial and memory-bound phases. The phase split at eight threads confirms the picture: the two parallel loops account for 1.92 ms of the 2.06 ms step, while the merge of the thread contributions and the serial interface solve account for the remaining 0.14 ms.

Ranks	Per step [ms]	Reduction [ms]	Speedup	Efficiency
sequential	8.07	–	–	–
2	4.66	0.09	1.73	0.87
4	4.29	0.71	1.88	0.47
8	2.56	0.20	3.15	0.39

Table 10: MPI strong scaling of the condensed step on the same mesh: medians over 200 steps and three runs. The reduction column is the median per-step cost of summing the interface system across the ranks (`MPI_Allreduce`).

Table 10 reports the MPI scaling. The single-rank run costs 8.93 ms against the 8.07 ms reference, an overhead of about eleven percent. Two ranks reach a speedup of 1.73, in line with the OpenMP result, but four ranks reach only 1.88: the median cost of the reduction grows to 0.71 ms and, since the reduction synchronises the ranks, it absorbs as waiting time any imbalance in the moment at which the ranks finish their elimination; the payload itself does not change. At eight ranks the reduction falls back to 0.20 ms and the speedup reaches 3.15. At every worker count the MPI version is slower than the OpenMP one on this machine, which is the expected outcome on a single shared-memory node: the MPI version pays the message and synchronisation costs and replicates the interface solve on every rank, without the shared address space that makes the threaded version cheap. These runs say nothing about distributed memory, where MPI is the only option of the two; we have not run the code on more than one node and draw no conclusion beyond one.

Figure 31(b) compares the four ways of arranging eight workers with the hybrid build, from one rank with eight threads to eight ranks with one thread each. The step time degrades monotonically as workers shift from threads to ranks, from 2.06 to 2.61 ms, and the two ends reproduce the pure OpenMP and pure MPI measurements within the variability across sessions; the reduction cost grows with the number of ranks, from essentially zero to 0.21 ms. On one

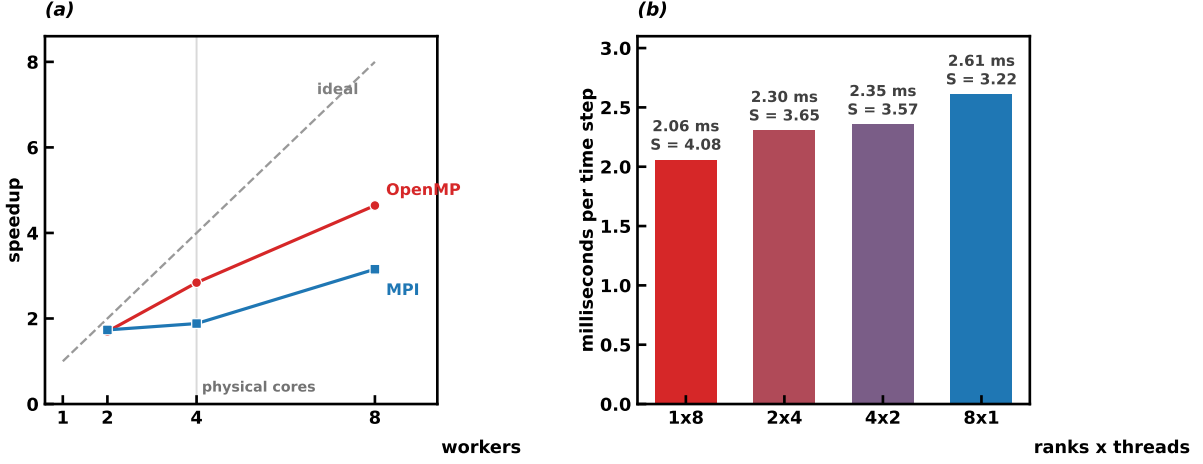


Figure 31: Parallel performance on the mesh with 128 elements per connection. (a) strong scaling of the OpenMP and MPI versions; each series is normalised by the sequential condensed reference of its own experiment (9.58 and 8.07 ms per step). The one-worker configurations, omitted from the curves, cost 10.70 and 8.93 ms. (b) the four arrangements of eight workers with the hybrid build, as MPI ranks $\times$ OpenMP threads, with speedups against the reference of the hybrid experiment (8.40 ms per step).

node, at a fixed number of workers, threads are preferable to ranks and the hybrid arrangements sit in between.

Combining the algorithmic and the parallel gains, and comparing measurements from different sessions, the cost of a time step on the 128-element mesh falls from 64.5 ms for the best full-system factorisation to 2.06 ms for the condensed solver on eight threads, a combined factor of about thirty; the condensation contributes a factor of 5.9 and the parallelisation the rest. The performance study thus closes the chain opened by Table 6. The anomaly of the sequential baseline was an ordering artefact and the natural ordering with a symmetric factorisation removes it; static condensation turns the arrow structure of the connectome discretisation into independent per-connection work with a constant-cost interface; OpenMP converts that structure into a speedup of 2.84 on the four physical cores and 4.64 with SMT; the MPI version is correct and scales, but on a single node it does not match the threaded version and its value lies in distributed settings we have not measured. Every parallel configuration reproduces the sequential trajectory to round-off, so all of the above is a reorganisation of the same computation.